




**«Approved»
by the Dean**
Syzdykova Z.A.
 « » of 2024

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	Programming for data analysis and database
Degree cycle (level)/major	MSc, Applied Data Analytics
Year, term	Year 1, Term 1
Number of credits	5
Language of delivery:	English, Russian
Prerequisites	There are no specific prerequisites or experience required for the course. However, it will be only beneficial if students have a solid understanding of programming, databases, and statistics / probability / linear algebra for the data science specialization.
Postrequisites	More advanced courses on Data Science, Machine Learning / Deep Learning, and Statistics as well as projects (including master's dissertation) / case studies related to students' major.
Lecturer(s)	Svitlana Biloshchytska, Doctor of Technical Sciences, Associate Professor Astana IT University, Expo, C1 block, 3d floor, office C1.1.336 bsv@astanait.edu.kz
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	“Programming for Data Science” is a 10-week course, which offers students the opportunity to learn the most important programming languages used by many data scientists worldwide. This is an introductory course for the first-year MSc students at Astana IT University. After completion, students will be proficient in the programming skills as well as state-of-the-art tools used in many data science projects, including SQL and Python. Having mastered these in-demand tools and techniques, students will be able to tackle real-world data analysis problems. Learning to program in Python and SQL is the core of this course. It also includes learning specialized data libraries for Python, such as Pandas, Numpy, learning how to visualize data, basics of statistic and probability and others. By learning these foundations, students will be ready to advance their career in my data related fields such data engineering, data analytics, data science, big data, business analytics and so on.
2. Course Goal(s)	The main goal of the course is to provide a solid background in programming in the field of data science and data analysis for MSc students of “Applied Data Analytics” program in preparation either for a job in industry or for more advanced courses at the graduate level.

3. Course Objectives:	Course objectives include but no limited to: - Learn main data scientists' toolkit; - Learn libraries and toolkits for NLP; - Understand how to work with relational databases using SQL queries; - Conceptualize data structures in Python; - Analyze how to work with data in Python using Pandas and Numpy libraries; - Improve understanding of how to make statistical experiments in Python; - Implement application interfaces and data collection principles in projects; - Know the foundation of data visualization.
4. Skills & Competences	Course skills and competences include but not limited to: - Working with latest tools for Data Science projects; - Understanding the main concepts of programming in Python; - Working with various data structures, classes, and objects in Python; - Using Python libraries for NLP for Data Science projects depending on the tasks; - Writing general and advanced SQL queries; - Using the most popular Python libraries for Data Science projects Pandas and Numpy depending on the given problems; - Understanding the 9 main statistical tools available in Python for Data Science projects; - Dealing with APIs and Data collection tools; - Applying data visualization tools and techniques in Data Science projects.
5. Course Learning Outcomes:	By the end of this course students will be able to: - Understand the main tools used by many data scientists worldwide; - Apply NLP to solve real data science problems; - Apply SQL and Python to solve real world data science problems; - Use various APIs and data collection techniques required for data science projects; - Implement data visualization in data science projects.
6. Methods of Assessment	Methods of assessment students' progress will be based on assignments (5 assignment per course), MidTerm, EndTerm and a Final examination. The further instruction will be provided in the course page in Moodle.
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). Available in the university library: - Database Systems: Design, Implementation, & Management 12th Edition, Carlos Coronel, Steven Morris, Cengage, 2016 - Data Science from Scratch: First Principles with Python 2nd Edition - Fundamentals of Probability and Statistics for Engineers Available on the internet for free:

	- Statistical Learning with Sparsity: The Lasso and Generalizations by Trevor Hastie, Robert Tibshirani, Martin Wainwright (can be downloaded here - https://web.stanford.edu/~hastie/StatLearnSparsity) - The Field Guide to Data Science by Booz Allen Hamilton (can be downloaded here - https://www.researchgate.net/publication/258698880_The_Field_Guide_to_Data_Science) - Data Mining and Analysis by Mohammed J. Zaki and Wagner Meira, Jr (can be read online here - https://dataminingbook.info/) - Data Science for Business by Tom Fawcett and Foster Provost (can be downloaded here - https://www.researchgate.net/publication/256438799_Data_Science_for_Business) - An Introduction to Data Science by Jeffrey Stanton, Syracuse University (can be downloaded here - https://docs.google.com/file/d/0B6iefdnF22XQeVZDSknjZ0Z5VUhzedit?pli=1) - Python for You and Me by Kushal Das (can be downloaded here - https://pymbook.readthedocs.io/en/latest/) - Python for Everybody by Dr. Charles R Severance (can be downloaded here - http://do1.dr-chuck.com/pythonlearn/EN_us/pythonlearn.pdf) - SQL Tutorial as a PDF (can be downloaded here - http://www.tutorialspoint.com/sql/sql_tutorial.pdf) - Structured Query Language (SQL): a Practical Introduction by Philip Greenspun (can be downloaded here - http://philip.greenspun.com/sql/)
8. Resources	Online journals, article, papers, books and internet resources as well as online emulators and online software for simulation.
9. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as “not graded”. In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean’s office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing.

Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties:

- Offers a different and unique, but relevant, perspective;
- Contributes to moving the discussion and analysis forward;
- Builds on other comments.

Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes.

Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor.

Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course.

Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.

Final exam*: The final exam for the course will be held in the form of a practical exam. More details will be provided at the time of final exam weeks. At the completion of the exam, all works must be submitted in the Learning Management System (moodle.astanait.edu.kz). No late submissions are allowed in the exam.

Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld

	devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. Online lessons can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. Cheating and plagiarism are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.
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* Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Introduction and Data Scientist's Toolkit - Introduction to the Course (course goal and objectives, course policies, methods of assessment, course learning outcomes); - Data Scientist's Toolkit (Python, R / R studio, SQL / No SQL, open source / commercial tools, cloud-based tools, libraries for Data Science, application interfaces, datasets, Jupyter notebook / kernels / architecture, Google Colab, Git / GitHub).	3	2		1	9
2	Python Fundamentals and Data Structures in Python - Python Basics (types, expressions and variables, string operations) - Python Data Structures (Lists and Tuples, Dictionaries, Sets) - Python Fundamentals (Conditions, Loops, Functions, Exception Handling, Objects and Classes) - Reading / Writing Files	3	2		1	9
3	Working with Data in Python: Numpy	3	2		1	9
4	Working with Data in Python: Pandas	3	2		1	9
5	Introduction to NLP - Libraries and toolkits for NLP - Language processing in information retrieval tasks. - Morphological analysis. - Automatic referencing. - Sentiment analysis. Information extraction.	3	2		1	9
6	Introduction to NLP - Automatic referencing. - Sentiment analysis. Information extraction.	3	2		1	9

7	Statistics for Data Science - Descriptive Statistics (types of data, descriptive statistics, central tendency, dispersion) - Probability Distribution (random numbers, probability distribution, normal distribution, t distribution) - Hypothesis testing (z-test, t-test, dealing with tails and rejections, anova, correlation tests) - Regression Analysis	3	2		1	9
8	Relational Databases and SQL	3	2		1	9
9	Data Visualization	3	2		1	9
10	APIs and Data Collection - Simple APIs / Rest APIs / HTTP Requests - Web scraping with Python - Working with different file formats (csv, xml, json, xlsx)	3	2		1	9
	Total hours: 150	30	20		10	90

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	Data Scientist's Toolkit (Python, R / R studio, SQL / No SQL, open source / commercial tools, cloud-based tools, libraries for Data Science, application interfaces, datasets, Jupyter notebook / kernels / architecture, Google Colab, Git / GitHub).	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
2	Python Basics (types, expressions and variables, string operations) Python Data Structures (Lists and Tuples, Dictionaries, Sets) Python Fundamentals (Conditions, Loops, Functions, Exception Handling, Objects and Classes) Reading / Writing Files	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
3	Working with Data in Python: Pandas	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
4	Working with Data in Python:	9	Reading list, books,	Reading list,

	Numpy		internet resources Electronic version	books, internet resources Electronic version
5	Libraries and toolkits for NLP Language processing in information retrieval tasks. Morphological analysis. Automatic referencing. Sentiment analysis. Information extraction.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
6	Automatic referencing. Sentiment analysis. Information extraction.	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
7	Descriptive Statistics (types of data, descriptive statistics, central tendency, dispersion) Probability Distribution (random numbers, probability distribution, normal distribution, t distribution) Hypothesis testing (z-test, t-test, dealing with tails and rejections, anova, correlation tests) Regression Analysis	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
8	Relational Databases and SQL	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
9	Data Visualization	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
10	Simple APIs / Rest APIs / HTTP Requests Web scraping with Python Working with different file formats (csv, xml, json, xlsx)	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Assignment 1 (by the end of Week 3) Assignment 2 (by the end of Week 5) Mid Term (Week 5)	30 30 40	100

2 nd attestation	Assignment 3 (by the end of Week 8) Assignment 4 (by the end of Week 10) End Term (Week 10)	30 30 40	100
Final exam*	Quiz		100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	
C-	1,67	60-64	Satisfactory
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement

60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Biloshchytska Svitlana	Professor	20/08/2024	

Director

 Sergaziyev M.Zh.